Protein–Ligand Binding Prediction

Machine Learning on the KIBA Dataset

1 000-protein benchmark · Cold-protein evaluation · 5 models

ESM-2 protein embeddings · Morgan + MACCS fingerprints · RDKit descriptors

Pipeline Overview

1 Data

KIBA kinase activity dataset · binder threshold < 12.1

→

2 Proteins

UniProt sequences → ESM-2 (650M) mean+max pooling dim = 960

→

3 Ligands

PubChem SMILES → Morgan+MACCS+ atom-pair FPs + RDKit dim=2229

→

4 Split

Cold-protein (no UniProt_ID overlap between train & test)

→

5 Models

Logistic Reg.
Random Forest
XGBoost · LightGBM
Interaction MLP

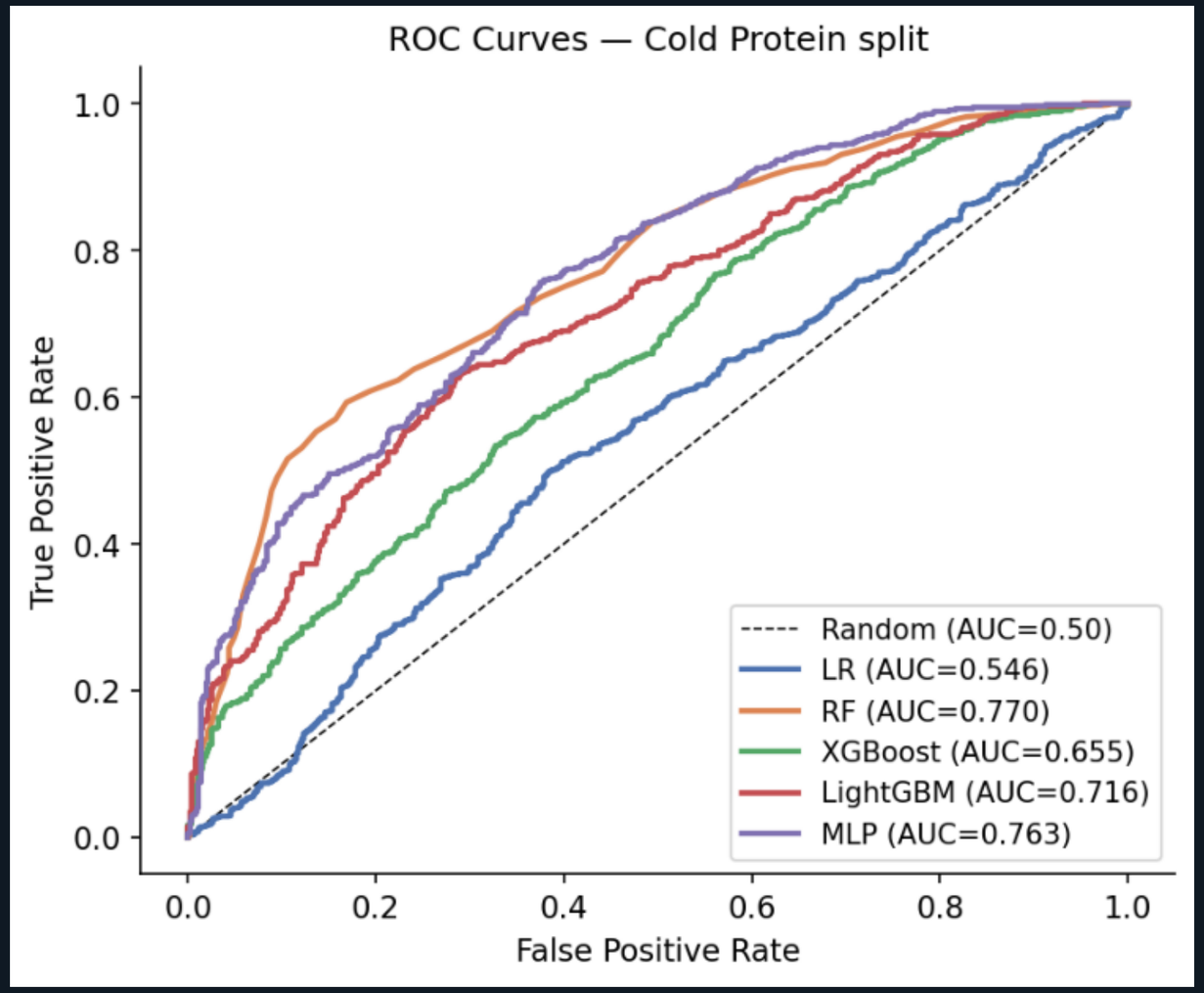
→

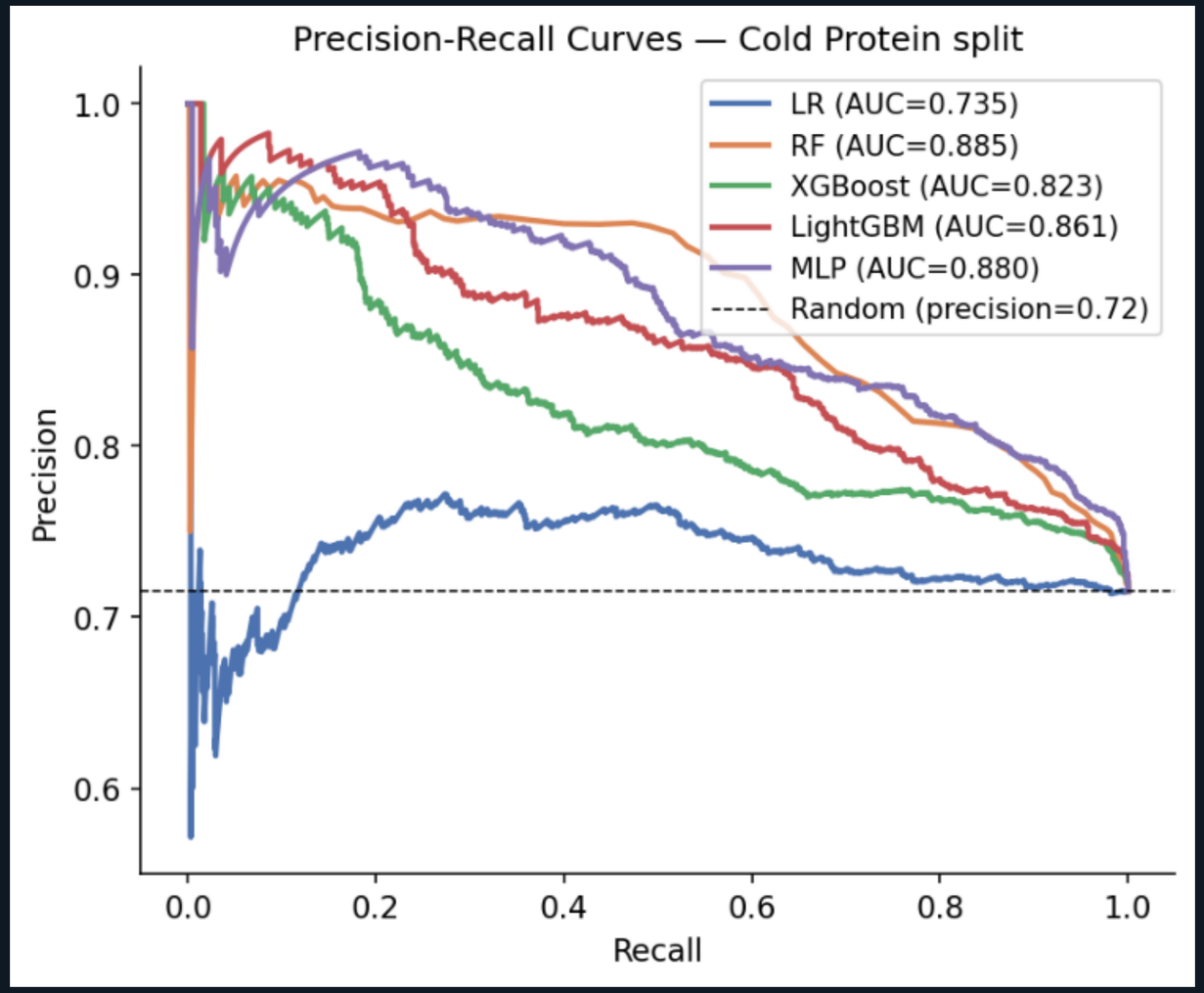
6 Evaluation

ROC-AUC · PR-AUC
BEDROC · EF@1%/5%
5-fold protein-aware CV

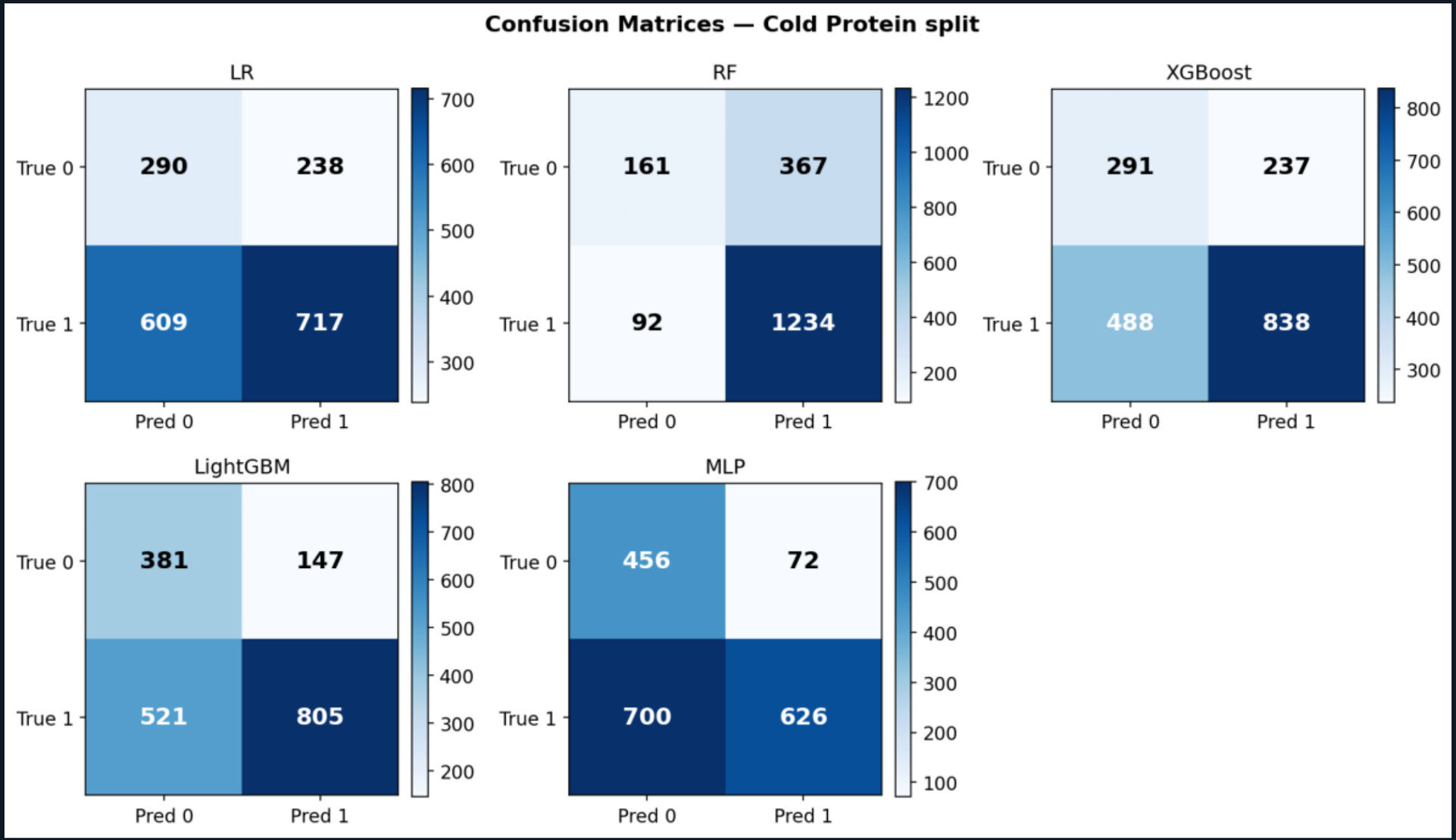
Model Comparison — Test Set (Cold-Protein Split)

ROC and PR Curves — Cold-Protein Test Set

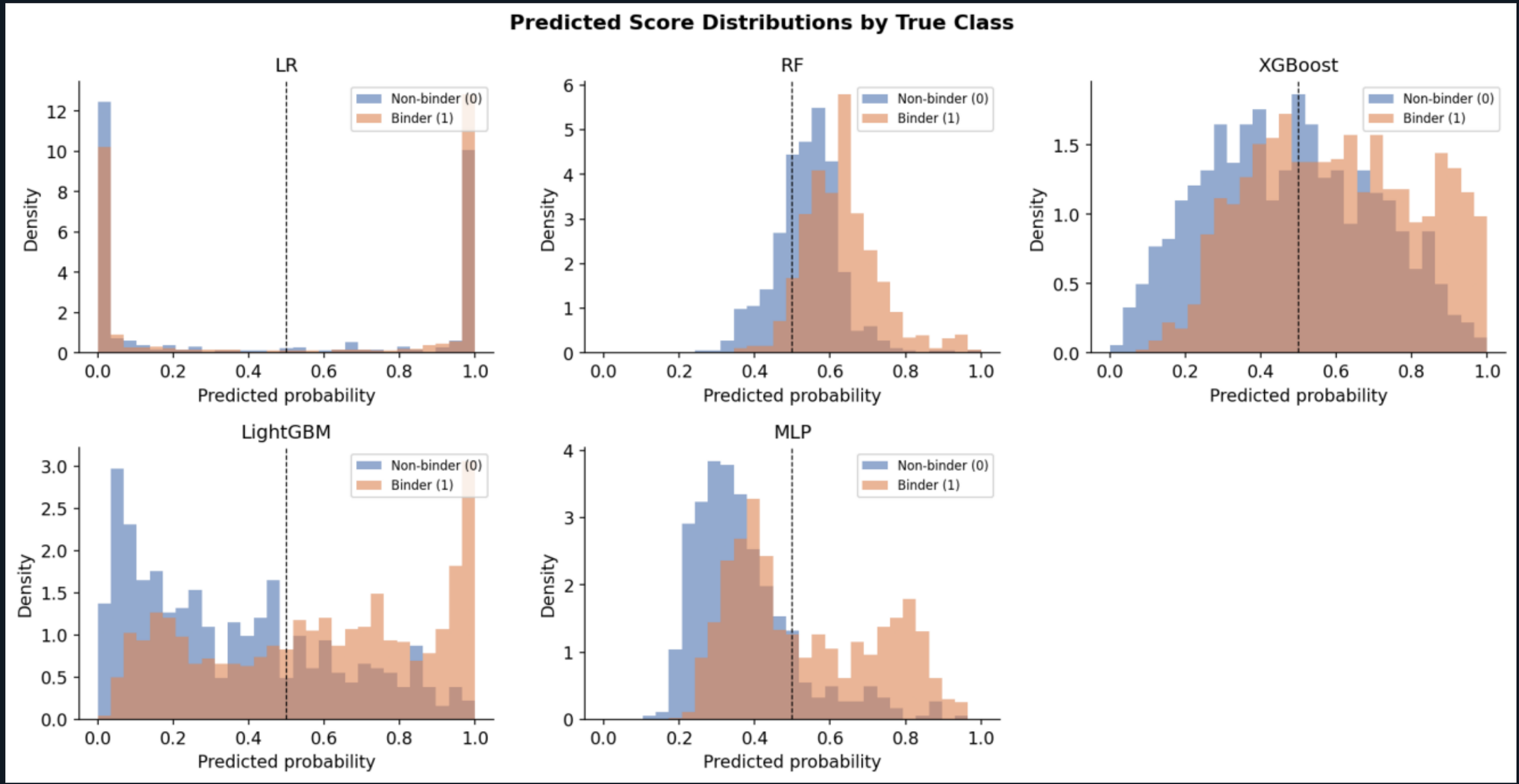




Confusion Matrices — Cold-Protein Test Set

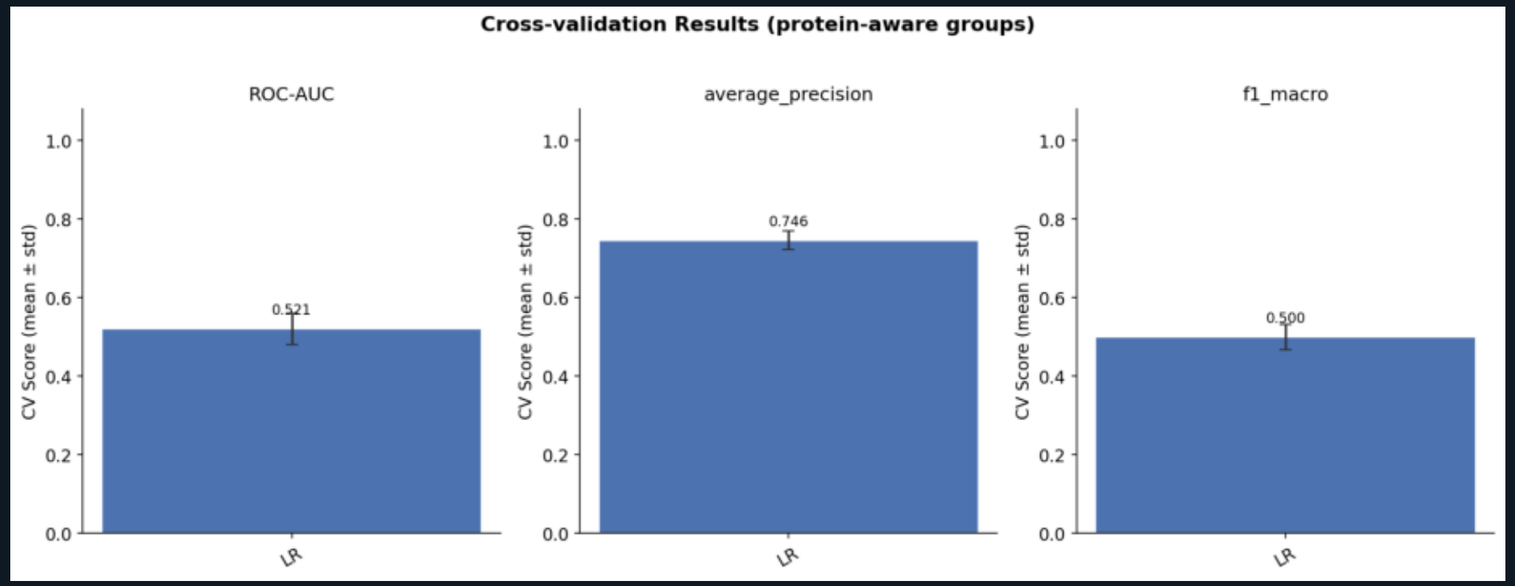


Predicted Score Distributions



5-Fold Protein-Aware Cross-Validation

StratifiedGroupKFold — no protein appears in both train and validation folds



Key Finding

CV ROC-AUC ≈ 0.52 (near chance)

Honest estimate for unseen proteins. Single-split 0.77 is optimistic — proteins leak between folds.

LR CV: 0.521 ± 0.040

Results Summary — All Models

Model	ROC-AUC	PR-AUC	F1	Accuracy	BEDROC	EF@1%
Logistic Reg.	0.5459	0.7359	0.6287	0.5431	0.7359	0.9567
Random Forest	0.7698	0.8833	0.8432	0.7524	0.8833	1.3246
XGBoost	0.6554	0.8237	0.6980	0.6090	0.8237	1.3982
LightGBM	0.7163	0.8614	0.7068	0.6397	0.8614	1.3246
MLP	0.7626	0.8802	0.6186	0.5836	0.8802	1.3246

Limitations & Next Steps

Current Limitations

- Cold-protein CV ROC-AUC ≈ 0.52 — generalisation
- to novel proteins is near chance-level
- KIBA scores are partially estimated (not all measured)
- Ligand features are fingerprint-based — no 3-D geometry
- No cross-docking or structure-aware features used

Next Steps

- Graph Neural Networks on molecular graphs (GNN-DTI)
- Structure-aware models: AlphaFold2 binding site features
- Multi-task learning across kinase families
- Active learning loop: prioritise wet-lab validation hits
- Larger benchmark: full KIBA + BindingDB + ChEMBL